Optimization for Deep Learning (OPT4DL)

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Beyond Convexity: (S)GD and Smoothness

Beyond Convex Functions



- So far, we discussed how GD and SGD perform if the function is convex
- Under Lipschitzness, both satisfy $\mathcal{O}(1/\sqrt{T})$ rate to find an approximate optimal solution
- In DL, however, the loss is nonconvex. Thus, we need to relax convexity
- Difficult goal! We thus also relax our notation of convergence to an approximate first-order solution

$$\mathbb{E}[f(\hat{w}) - f(w^*)] \leq \epsilon \Rightarrow \|\nabla f(\hat{w})\|^2 \leq \epsilon \tag{1}$$

- This condition requires the existences of gradient. Thus, instead of Lipschitzness we require another notion of functions' sensitivity called smoothness

Smooth functions



Definition

A function $f : \mathbb{R}^d\mathbb{R}$ is L-smooth if its gradient is L-Lipschitz:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \tag{2}$$

for all x, y and some constant $0 \leq L < \infty$.

- Recall that G-Lipschitzness, i.e. $|f(x) - f(y)| \leq G\|x - y\|$ implied the slope of f is bounded by G.
- Smoothness then means the slope of the ∇f, i.e., the second derivative of Hessian is bounded by L.
- More formally, if the function is twice differentiable, all eigen values of the Hessians are bounded:

$$-L \leq \lambda_d := \lambda_{min}(\nabla^2 f(x)), \ldots, \lambda_1 := \lambda_{max}(\nabla^2 f(x)) \leq L \tag{3}$$

for all x.

Examples



- $f(x) = \frac{1}{2} x^\top A x$

- $f(x) = e^x$

Smoothness and Bounded Curvature



- The condition $|\lambda_i(\nabla^2 f)| \leq L$, $i = 1, \dots d$ means the function has bounded curvature from above and from below
- That is, we can upper bound the function by a convex quadratic function and lower bound the function by a concave quadratic function at any point!
- Leading to the following sometimes more useful equivalent definition of smoothness

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2 \quad \text{RHS a convex quadratic function of } x \tag{4}$$

$$f(x) + \langle \nabla f(x), y - x \rangle - \frac{L}{2}\|y - x\|^2 \leq f(y) \quad \text{LHS a concave quadratic function of } x \tag{5}$$

Let $\Delta := f(w_1) - f(w^*)$ denote the initial suboptimality.

Theorem (GD)

Assume f is L-smooth. then, GD with $\eta = \frac{1}{L}$ satisfies

$$\frac{1}{T}\sum_{t=1}^{T} \|\nabla f(w_t)\|^2 \leq \frac{2L\Delta}{T} = \mathcal{O}(1/T). \tag{6}$$

Theorem (SGD)

Assume f is L-smooth. Furthermore assume we have access to an SFO and that $z_1, \ldots, z_T$ are statistically independent. Then, SGD with learning rate $\eta = \min\{\frac{1}{L}, \sqrt{\frac{2\Delta}{L\sigma^2 T}}\}$ satisfies

$$\mathbb{E}_{z_1,\ldots,z_T}[\frac{1}{T}\sum_{t=1}^{T} \|\nabla f(w_t)\|^2] \leq \frac{2L\Delta}{T} + 2\sigma\sqrt{\frac{2\Delta L}{T}} = \mathcal{O}(1/\sqrt{T}). \tag{7}$$

(S)GD and Smooth functions (Cont'd)



- Dimension-free result
- Regulating role of η
- We can use either of 2 approaches to generate an output, that is (i) the last/best parameter w_T and (ii) sampling one iterate at random. The later implies $\mathbb{E}_{\hat{w}}\|\nabla f(\hat{w})\|^2 \leq \frac{2L\Delta}{T}$ for GD and $\mathbb{E}_{\hat{w}}\|\nabla f(\hat{w})\|^2 \leq \frac{2L\Delta}{T} + 2\sigma\sqrt{\frac{2\Delta L}{T}}$ for SGD
- Iteration complexity of $T = \Omega(\epsilon^{-1})$ for GD and $T = \Omega(\epsilon^{-2})$ for SGD
- SGD bound consist of a higher order term (first term matching GD's) and the dominant term (second term due to stochasticity)
- Iteration complexity of $T = \Omega(\epsilon^{-2})$ compared to $T = \Omega(\epsilon^{-1})$ for GD
- Bound tight in general, but with more structures (present in DL optimization) we can show the theoretical of benefits of advance training algorithms.

GD: A new Perspective



- So far we motivated GD from the fixed point calculation and following the direction of descent locally perspectives
- Thanks to smoothness, we can develop anew perspective which also gives us a way to develop better/generalized methods
- Consider minimizing $f(w)$ iteratively. It is desired to update the parameter such that we always make a progress:

$$f(w_{t+1}) \leq f(w_t), \qquad \forall t \tag{8}$$

  That is our method has the descent property.
- Going one step further, maximizing the progress is preferred:

$$\max_{w_{t+1}} P_t := f(w_t) - f(w_{t+1}) \tag{9}$$

- A general recipe to do so is called Majorization-Minimization when minimizing loss/cost, or equivalently Minorization-Maximization if maximizing profit/utility/reward

Majorization-Minimization (MM)



- Maximize the P_t as much as possible. We will do so through a surrogate upperbound U_t with the following properties
  - $U_t(w_t) = f(w_t)$ (surrogate is tight at the current solution)
  - $U_t(w) \geq f(w)$ (surrogate is an upperbound on the loss function)
  - $U_t(\cdot)$ is simple to minimize
- Thus, our MM scheme is as follows
  - $w_{t+1} = \arg\min_w U_t(w)$
  - Update the surrogate to go from U_t to the new surrogate U_{t+1} satisfying the above properties
- Examples include K-means clustering and Expectation-Maximization (EM) for solving mixture models.
- We will show GD is also an example of MM for smooth functions

Why those properties?



- The properties of the surrogate function guarantee progress

$$
\begin{aligned}
P_t = f(w_t) - f(w_{t+1}) &\geq f(w_t) - U_t(w_{t+1}) && \text{upper bound property} \\
&\geq f(w_t) - U_t(w_t) && \text{parameters are minimizers of the surrogate} \\
&\geq f(w_t) - f(w_t) = 0 && \text{tightness property}
\end{aligned} \tag{10}
$$

- This in addition to assuming the loss is bounded from below guarantees the method will converge/stop at a stationary point.

GD as MM



- Recall by smoothness we have a convex quadratic upperbound on the loss!
- This could be our surrogate:

$$U_t(w) := f(w_t) + \langle w - w_t, \nabla f(w_t) \rangle + \frac{L}{2}\|w - w_t\|^2 \tag{11}$$

- It is tight at w_t
- It is always an upper bound on f by smoothness
- It is strongly convex (easy to optimize). Just take the gradient and set it equal to zero:

$$\begin{aligned} &\nabla U_t(w) = 0 \\ &0 + \nabla f(w_t) + L(w - w_t) = 0 \\ &w = w_t - \frac{1}{L}\nabla f(w_t) \end{aligned} \tag{12}$$

- This is exactly GD with learning rate $\eta = 1/L$
- More generally, GD with $\eta \leq 1/L$ is an instance of MM



- A procedure to handle constraints (e.g. Projected GD: $w_{t+1} = \mathcal{P}_{\mathcal{X}}(w_t - \eta \nabla f(w_t))$)

$$w_{t+1} = \arg\min_{w \in \mathcal{X}} f(w_t) + \langle w - w_t, \nabla f(w_t) \rangle + \frac{1}{2\eta} \|w - w_t\|^2 \tag{13}$$

- Extension to broad distance/divergence notions (e.g. Mirror Descent)

$$w_{t+1} = \arg\min_{w \in \mathcal{X}} f(w_t) + \langle w - w_t, \nabla f(w_t) \rangle + \frac{1}{2\eta} \mathcal{D}(w, w_t) \tag{14}$$

  Useful when w is a probability vector, e.g. reinforcement learning.

- Extension to arbitrary update vectors g_t

$$w_{t+1} = \arg\min_{w \in \mathcal{X}} f(w_t) + \langle w - w_t, g_t \rangle + \frac{1}{2\eta} \mathcal{D}(w, w_t) \tag{15}$$

  Results in Stochastic (Projected) Gradient Descent, Stochastic Mirror Descent, etc.

- Possible to find better surrogates if f has more structures (e.g., leveraging Hessian's properties).

Proof of GD's Theorem



- Study the amount of progress while leveraging smoothness

$$f(w_{t+1}) \leq f(w_t) + \langle w_{t+1} - w_t, \nabla f(w_t) \rangle + \frac{L}{2}\|w_{t+1} - w_t\|^2 \tag{16}$$

- Using GD's update $w_{t+1} = w_t - \eta \nabla f(w_t)$

$$\begin{aligned} f(w_{t+1}) &\leq f(w_t) - \eta \langle \nabla f(w_t), \nabla f(w_t) \rangle + \frac{L\eta^2}{2}\|\nabla f(w_t)\|^2 \\ &= f(w_t) - \|\nabla f(w_t)\|^2 \eta (1 - \frac{L\eta}{2}) \\ \Rightarrow P_t &= f(w_t) - f(w_{t+1}) = \|\nabla f(w_t)\|^2 \eta (1 - \frac{L\eta}{2}) \end{aligned} \tag{17}$$

- Side note 1: the larger the gradient, the more progress we make: $P_t = \mathcal{O}(\|\nabla f(w_t)\|^2)$
- Side note 2: for any $\eta < 2/L$ we make progress.
- Side note 3: Maximum progress when $\eta = 1/L$ (progress a concave quadratic function of η).

Proof of GD's Theorem (Cont'd)



- Rearrrange and average over t (requires $\eta < 2/L$)

$$\frac{1}{T}\sum_{t=1}^{T}\|\nabla f(w_t)\|^2 \leq \frac{\sum_{t=1}^{T} f(w_t) - f(w_{t+1})}{T\eta(1-\frac{L\eta}{2})} \tag{18}$$

- Telescoping structure and noting $f(w_1) - f(w_{t+1}) \leq f(w_1) - f(w^*) = \Delta$

$$\frac{1}{T}\sum_{t=1}^{T}\|\nabla f(w_t)\|^2 \leq \frac{\Delta}{T\eta(1-\frac{L\eta}{2})} \tag{19}$$

- Optimizing the bound w.r.t. η gives $\eta = 1/L$ and the stated result.

Proof of SGD's Theorem



- Following a similar approach we use smoothness and SGD's update to study progress

$$\begin{aligned} f(w_{t+1}) &\leq f(w_t) + \langle w_{t+1} - w_t, \nabla f(w_t) \rangle + \frac{L}{2}\|w_{t+1} - w_t\|^2 \\ &= f(w_t) - \eta \langle g_t, \nabla f(w_t) \rangle + \frac{L\eta^2}{2}\|g_t\|^2 \end{aligned} \tag{20}$$

  but here w_t's and g_t's are random quantities. We need proper expectations (hence the SFO and independent assumptions)

- First, conditioned on w_t (equivalently $z_1, \ldots, z_{t-1}$)

$$\mathbb{E}_{z_t}[\langle g_t, \nabla f(w_t) \rangle | w_t] = \langle \mathbb{E}_{z_t}[g_t | w_t], \nabla f(w_t) \rangle = \|\nabla f(w_t)\|^2 \tag{21}$$

- Second, conditioned on w_t (equivalently $z_1, \ldots, z_{t-1}$)

$$\begin{aligned} \mathbb{E}_{z_t}[\|g_t\|^2 | w_t] &= \mathbb{E}_{z_t}[\|g_t \pm \nabla f(w_t)\|^2 | w_t] \\ &= \mathbb{E}_{z_t}[\|g_t - \nabla f(w_t)\|^2 | w_t] + \|\nabla f(w_t)\|^2 \leq \|\nabla f(w_t)\|^2 + \sigma^2 \end{aligned} \tag{22}$$



- Thus, conditioned on w_t (equivalently $z_1, \ldots, z_{t-1}$)

$$\mathbb{E}_{z_t}[f(w_{t+1})|w_t] \leq f(w_t) - \|\nabla f(w_t)\|^2 \eta(1 - \frac{L\eta}{2}) + \frac{L\eta^2\sigma^2}{2} \tag{23}$$

  Side note: No guarantee on the (expected) progress anymore!

- Averaging over t, rearranging (requires $\eta < 2/L$), and taking expectation w.r.t. $z_1, \ldots, z_T$

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}_{z_{1:T}} \|\nabla f(w_t)\|^2 \leq \frac{\sum_{t=1}^{T} f(w_t) - f(w_{t+1})}{T\eta(1 - \frac{L\eta}{2})} + \frac{L\eta\sigma^2}{2(1 - \frac{L\eta}{2})} \tag{24}$$

- Telescoping structure and noting $f(w_1) - f(w_{t+1}) \leq f(w_1) - f(w^*) = \Delta$

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}_{z_{1:T}} \|\nabla f(w_t)\|^2 \leq \frac{\Delta}{T\eta(1 - \frac{L\eta}{2})} + \frac{L\eta\sigma^2}{2(1 - \frac{L\eta}{2})} \tag{25}$$

- Optimizing the bound w.r.t. η gives the stated result (doable but a bit cumbersome).